

Computer Vision Image Classification using ResNet50

Project Overview

This project demonstrates the development of an image classification model capable of recognizing ten object categories from the CIFAR-10 dataset using transfer learning and deep learning techniques.

The objective was not only to achieve accurate predictions, but also to gain practical experience with the complete computer vision workflow, including data preprocessing, model training, performance evaluation, and result interpretation.

The project was completed as part of the Data Analytics & Artificial Intelligence Program at Masterschool.

Dataset

The CIFAR-10 dataset contains 60,000 color images (32×32 pixels) distributed across 10 classes:

- Airplane
- Automobile
- Bird
- Cat
- Deer
- Dog
- Frog
- Horse
- Ship
- Truck

For computational efficiency, a subset of 10,000 images was used during model development and training.

Methodology

Data Preparation

- Image normalization
- Train-validation-test split
- One-hot encoding of class labels
- Data pipeline optimization

Model Architecture

The model was built using Transfer Learning with ResNet50 pretrained on ImageNet.

Architecture:

Input Image



ResNet50 Backbone (Pretrained)



Global Average Pooling



Dense Layer (128 neurons, ReLU)



Dropout (0.3)



Dense Layer (64 neurons, ReLU)



Dropout (0.3)



Output Layer (10 classes, Softmax)

Training Strategy

Stage 1:

- Frozen ResNet50 feature extractor
- Training of classification layers

Stage 2:

- Fine-tuning of selected ResNet50 layers
 - Lower learning rate
 - Early stopping to prevent overfitting
-

Technologies Used

Programming

- Python

Deep Learning

- TensorFlow
- Keras

Data Analysis

- NumPy
- Pandas

Visualization

- Matplotlib
- Seaborn

Environment

- Google Colab
-

Results

Final Model Performance:

Metric	Value
Test Accuracy	65.03%
Test Loss	1.02

Key Findings

- Transfer learning significantly improved performance compared with training a model from scratch.
- Fine-tuning enabled better feature extraction and improved classification accuracy.
- The model performed well on visually distinct classes.
- Most classification errors occurred between visually similar categories such as:

- Cat vs Dog
- Deer vs Horse
- Automobile vs Truck

Example Workflow

Data Loading



Preprocessing



Transfer Learning



Model Training



Fine-Tuning



Performance Evaluation



Prediction Analysis

Skills Demonstrated

- Computer Vision
- Deep Learning
- Transfer Learning
- Image Classification
- Neural Networks
- Data Preprocessing
- Model Evaluation
- Python Programming
- Scientific Data Analysis
- Experimental Design

Future Improvements

Potential directions for improving model performance include:

- Using the complete CIFAR-10 dataset
- Advanced data augmentation
- Hyperparameter optimization
- Higher image resolution datasets
- Exploring alternative architectures such as:
 - EfficientNet
 - DenseNet
 - Vision Transformers (ViT)

Author

Dr. Ouad Soltani

Plant Scientist | Data Analytics & Artificial Intelligence